

MacSync-WX1-PO RS485 & Analog to Wi-Fi Data Logger Datasheet

Product Overview

MacSync-WX1-PO is an industrial-grade **RS485 & Analog to Wi-Fi data logger**, designed for reliable data acquisition and real-time communication in industrial and commercial environments where Wi-Fi connectivity is available. It retains the proven hardware architecture of the MacSync data logger family with **direct IP-based Wi-Fi connectivity**.

The device seamlessly interfaces with **MODBUS RTU sensors, energy meters, PLCs, transmitters, and industrial instruments**, acquiring data from **RS485, 4–20 mA, 0–10 V, and digital inputs**. Collected data is securely transmitted to **on-premise servers, local dashboards, or cloud platforms** using standard protocols such as **MQTT, HTTP/HTTPS, and TCP/IP**, without requiring any external gateways.

Built for continuous 24×7 operation, MacSync-WX1-PO features a robust power-operated design, noise-immune circuitry, and an industrial enclosure suitable for factories, utilities, buildings, cold storage, and infrastructure monitoring. **Onboard data buffering and store-and-forward mechanisms** ensure data integrity even during temporary network or server outages.

Whether used for **industrial automation, energy monitoring, building management systems, process monitoring, or equipment health tracking**, MacSync-WX1-PO delivers **fast, reliable, and scalable Wi-Fi-based data logging**, making it ideal for deployments that demand low latency, local control, and seamless IT/OT integration.





Key Features

RS485 (MODBUS RTU) Interface

Seamlessly connects with industrial controllers, meters, and smart instruments.

Compatible with Energy Meters, PLCs & Transmitters

Designed for universal industrial integration.

Robust & Weatherproof

Built with an industrial-grade enclosure suitable for harsh environments.

Designed for Industrial & Outdoor Deployments

Perfect for factories, utilities, agriculture, and remote monitoring.

Macnman MacTalk Protocol Support

Enables device-to-device LoRa communication without a gateway.

Direct Control & Local Automation

Supports local device-to-device actions without cloud dependency.

Fully LoRaWAN® Compatible

Works with all network servers and standard LoRaWAN gateways.

4–20 mA, 0–10 V & Digital Inputs

Supports industry-standard analog sensors and digital switches.

Multiple Analog Channels

Enables simultaneous monitoring of multiple process parameters.

Smart Uplink Modes

Supports periodic, threshold-based, and event-driven LoRaWAN uplinks.

Easy BLE Configuration

Local setup and diagnostics made simple with the Maya mobile app.

Secure Onboarding

Ships with pre-flashed DevEUI, AppEUI, and AppKey for fast activation.

Optional Onboard Data Buffering

Prevents data loss during network downtime by storing records locally.

Suited Applications

Industrial Automation & SCADA

Logs data from PLCs, RTUs, VFDs, and industrial controllers over RS485 and publishes it to local SCADA or cloud platforms via Wi-Fi.

Process & Equipment Monitoring

Captures data from pressure, flow, level, and temperature transmitters using 4–20 mA and 0–10 V interfaces.

Cold Storage, HVAC & Utilities

Monitors temperature, humidity, compressors, and HVAC systems with low-latency Wi-Fi connectivity.

Data Centers & Server Rooms

Logs power, energy, temperature, humidity, and equipment health data with direct local network integration.

On-Premise Industrial IoT Deployments

Ideal for enterprises requiring local data ownership, low latency, and IT/OT integration without public networks.

Energy & Utility Metering

Integrates with RS485 energy, water, gas, and steam meters for real-time monitoring and on-premise or cloud-based analytics.

Factories & Manufacturing Plants

Enables machine data logging, production monitoring, and downtime analysis within Wi-Fi-enabled factory environments.

Building Automation Systems (BMS)

Connects meters, controllers, and sensors to BMS, IBMS, and facility dashboards without requiring gateways.

Warehouses & Logistics Facilities

Supports monitoring of power, environment, and equipment across large indoor facilities using existing Wi-Fi networks.

Technical Specifications

General Specifications

Part Order Number	MacSync-WX2-PO
Product Brand	Macnman Technologies Pvt Ltd

Interfaces Available

RS485 Interface	1 (Switchable)
4-20 mA Analog Channels	x 2 (Switchable)
0-10 V Analog Channels	x 2 (Switchable)
Digital Inputs	x 2 (Switchable)
ADC Resolution	12-bit

Note

The device use same channels for the RS485 & Analog inputs, either can be used at a time.

Wireless Communication

Wi-Fi Standards	IEEE 802.11 b/g/n/ac
Frequency Bands	2.4 GHz & 5 GHz (Dual-Band)
Operating Modes	Station Mode (Client) Access Point Mode (Configuration)
Network Protocols	MQTT, HTTP, HTTPS
Security	WPA2-PSK / WPA3 TLS / SSL Encrypted Communication
IP Configuration	DHCP / Static IP
Typical Indoor Range	Up to 100 m (environment dependent)

LED Status

LED Type	1 Internal Multi-Color LED
Solid Blue	Device is connected with Maya App
Flashing Blue	OTA (Over-the-Air) update in progress
Slow Blinking Green	Attempting to connect to the network
Single Red Blink	A confirmed parameter of MacSync is being modified

Device Configuration

Configuration	Via Macnman Maya Android App
Uplink Modes	Time Based / Threshold Based / Sampling Based
Stored Records*	30,000 Records
Remote Configuration	JSON Format Data - from server

Supported Cloud & Platforms

Deployment Models	On-Premise Servers Private Cloud Deployments
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	Public Cloud Platforms
IoT Cloud Platforms	AWS IoT Core Microsoft Azure IoT Hub Google Cloud (via MQTT / HTTPS) Custom IoT Platforms
Integration Methods	MQTT Brokers REST / HTTPS APIs Webhooks
Enterprise Systems	SCADA Systems BMS / IBMS Platforms Data Lakes & Analytics Engines

Physical Characteristics

Power Supply	9-36 V DC
Operating Temperature	-40°C to +80°C
Humidity Range	0% to 100% RH (non-condensing)
Ingress Protection	IP65 standard (IP67 optional variant)
Housing Material	Glass filled Nylon
Color	Matte White
Weight	~110g
Mounting Options	Pole Mounting / Wall Mounting

Electrical Characteristics

Input Voltage Range	9-36 V DC (External Supply)
Power Consumption	~100 µA in sleep mode ~150 mA during transmission
Input Impedance	>10 MΩ for voltage inputs

Communication Options

BLE Connectivity	For local configuration and diagnostics via Maya App
Cloud Protocols	MQTT / HTTPS
Serial Interface	RS485 Modbus RTU support (optional)

Certifications & Compliance

WPC	Compliant
RoHS	Conforms to RoHS directive

Security & Encryption

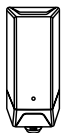
Data Encryption	AES-256 end-to-end encryption
Authentication	OTAA activation modes
Firmware Protection	Secure boot and signed OTA updates

MacSync-WX1-PO Variants

VARIANT NAME	POWER SOURCE	BATTERY CAPACITY	SOLAR SUPPORT	EXTERNAL FLASH	IDEAL FOR
MacSync-WX1-PO	12–24 V (Power)				Industrial or commercial sites with stable power supply
MacSync-WX1-BO	Battery	19 Ah			Remote or portable deployments with low power
MacSync-L-STD-SO-X1	Solar + Battery	12 Ah			Outdoor, off-grid, and energy-autonomous environments

What's in the Box?

Make sure box contains following items before installations



MacSync



**Mounting
Clip**



**User manual &
Warranty Card**



**Junction
Box***



**Mounting
Screws**

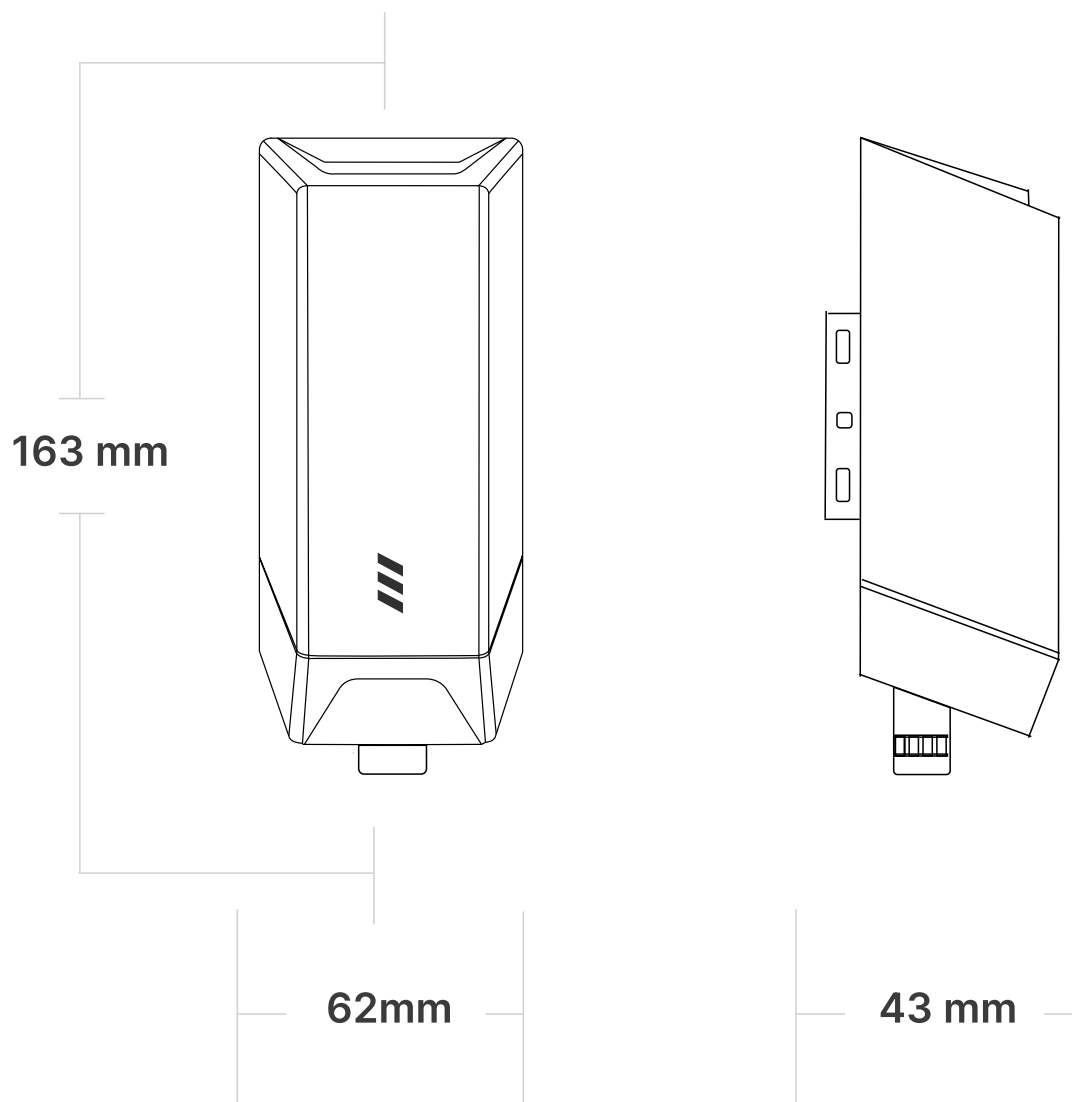


**Power
Adapters***

** Note**

Accessories must be purchased separately & may not be included in the box.

MacSync-WX1-PO Dimensions



Frequently Asked Questions

What is the MacSync-WX1-PO used for?

MacSync-WX1-PO is an industrial RS485 and analog to Wi-Fi data logger for reliable Modbus RTU and 4-20 mA sensor acquisition with real-time cloud communication. Typical applications include Industrial Automation & SCADA, Energy & Utility Metering, Process & Equipment Monitoring, Factories & Manufacturing Plants, Cold Storage, HVAC & Utilities and Building Automation Systems (BMS).

What inputs and outputs does the MacSync-WX1-PO have?

The MacSync-WX1-PO provides the following interfaces:

- RS485 Interface: 1 (Switchable)
- 4-20 mA Analog Channels: x 2 (Switchable)
- 0-10 V Analog Channels: x 2 (Switchable)
- Digital Inputs: x 2 (Switchable)
- ADC Resolution: 12-bit

How does the MacSync-WX1-PO connect and send data over Wi-Fi?

The MacSync-WX1-PO connects over IEEE 802.11 b/g/n/ac Wi-Fi on 2.4 GHz & 5 GHz (Dual-Band).

Security: WPA2-PSK / WPA3, TLS / SSL Encrypted Communication. IP configuration: DHCP / Static IP. It sends data using MQTT, HTTP, HTTPS. Typical indoor range is up to 100 m (environment dependent).

How is the MacSync-WX1-PO powered?

The MacSync-WX1-PO is powered from 9-36 V DC (External Supply).

What is the IP rating and operating temperature of the MacSync-WX1-PO?

The MacSync-WX1-PO has an ingress protection rating of IP65 standard (IP67 optional variant) and an operating temperature range of -40°C to +80°C. Operating humidity: 0% to 100% RH (non-condensing). Mounting options: Pole Mounting / Wall Mounting.

How is the MacSync-WX1-PO configured?

The MacSync-WX1-PO is configured with the Macnman Maya Android app. Supported uplink modes: Time Based / Threshold Based / Sampling Based. It can also be configured remotely with JSON data sent from the server.

Which cloud platforms does the MacSync-WX1-PO work with?

The MacSync-WX1-PO can send data to platforms including AWS IoT Core, Microsoft Azure IoT Hub, Google Cloud (via MQTT / HTTPS), Custom IoT Platforms. Deployment models: On-Premise Servers, Private Cloud Deployments, Public Cloud Platforms. Integration methods: MQTT Brokers, REST / HTTPS APIs, Webhooks.

Related Products and Resources

- User manual: [MacSync Wi-Fi datalogger user manual](#)
- Related datasheets: [MacSync LX1 RS485 Modbus RTU and analog to LoRaWAN datalogger](#), [MacSync-WX1-BO battery-operated RS485 and analog to Wi-Fi data logger](#)
- Product page: [MacSync-WX1-PO on macnman.com](#)
- Help: [Warranty terms and conditions](#) and [contact Macnman support](#)